

Exercise Sheet 2 - λ -calculus basics

Assigned: Monday 7 October
Suggested Deadline: Monday 14 October

Questions marked with * are optional, more difficult challenges.

Question 1: Parsing lambda terms

Draw the syntax trees for the following λ -terms:

- (a) $\lambda f. \lambda x. f\ x$
- (b) $\lambda x. \lambda y. x$
- (c) $(\lambda x. \lambda y. \lambda z. xy)\ z$
- (d) $(\lambda x. xx)(\lambda x. xx)$

Question 2: Free variables and substitution

- (a) Calculate the free variables of the following λ -terms. (Note that everything other than the λ symbol is a “variable” in the pure λ -calculus.) Try to be systematic in deriving your answer. Do not guess it directly!

- i. $\lambda x. \lambda y. x\ z$
- ii. $\lambda r. f\ r\ 1$
- iii. $\lambda y. * y\ 2$
- iv. $\lambda y. \lambda z. g\ (x\ z)\ (y\ z)$
- v. $\lambda x. \lambda x. x$

- (b) Write down the results of the following capture-avoiding substitutions.

- i. $(\lambda y. \lambda z. g\ (x\ z)\ (y\ z))\ [g \setminus (\lambda r. f\ r\ 1)]$
- ii. $(\lambda x. \lambda y. x\ z)\ [z \setminus (\lambda y. 2)]$
- iii. $(\lambda x. \lambda x. x)\ [x \setminus (\lambda y. * y\ 2)]$
- iv. $(\lambda x. \lambda y. x\ z)\ [z \setminus (y\ 2)]$

Question 3: beta reduction

- (a) Consider the lambda term $(\lambda f. \lambda x. f(fx)) (\lambda y. * y 2) 5$.
- Draw its abstract syntax tree.
 - β -reduce the term as far as possible.
- (b) Let S be the term $\lambda x. \lambda y. \lambda z. (xz)(yz)$ and K the term $\lambda x. \lambda y. x$. Reduce SKK to normal form. (Hint: This can be messy if you are not careful. Keep the abbreviations S and K around as long as you can and replace them with their corresponding λ -terms only if you need to. It becomes very easy then.)

Question 4: beta redexes

Given below is a term M :

$$M = \lambda x. ((\lambda z. z x)((\lambda r. \lambda s. s r) y f))$$

- Draw the syntax tree for M .
- Identify two separate beta redexes in the term M .
- Show the results of reducing M at each of these redexes. Call the results of these reductions N_1 and N_2 .
- Reduce N_1 and N_2 to normal forms and check if they reduce to the same normal form.
- A redex is said to be *outermost* if it is not a subterm of any other redex. Similarly, it is said to be *innermost* if it does not have a redex as a subterm. Specify if your redexes in step b are outermost or innermost redexes.
- What are the pro's and con's of always choosing outermost redexes for beta reduction? Similarly, for innermost redexes?
- Show that the term M above is β -equivalent to the following term:

$$N = \lambda x. ((\lambda z. \lambda u. \lambda v. u v z) x f y)$$

Question 5: Programming with untyped λ -calculus*

Recall the definitions from lectures of the λ -terms used to encode Booleans, Church numerals, and various functions. You may assume that those definitions behave as expected, e.g. that $\text{add } \underline{m} \ \underline{n} =_{\beta} \underline{m+n}$, etc.

- (a) Find a λ -term `iszero` such that

$$\text{iszero } \underline{n+1} =_{\beta} \text{F}$$

for all $n \in \mathbb{N}$ and $\text{iszero } \underline{0} =_{\beta} \text{T}$.

- (b) Defining

$$\text{pred} := \lambda n. n (\lambda p. \lambda b. b (p \text{F}) (\text{succ } (p \text{F}))) (\lambda b. b \underline{0} \underline{0}) \text{T}$$

prove that `pred` satisfies

$$\text{pred } \underline{n+1} =_{\beta} \underline{n}$$

for all $n \in \mathbb{N}$, as well as $\text{pred } \underline{0} =_{\beta} \underline{0}$.

(c) Hence find a λ -term **triangle** such that

$$\mathbf{triangle} \underline{n} =_{\beta} \underline{\frac{1}{2}n(n+1)}$$

for all $n \in \mathbb{N}$. (Recall that $\frac{1}{2}n(n+1) = 0 + 1 + \dots + n$).